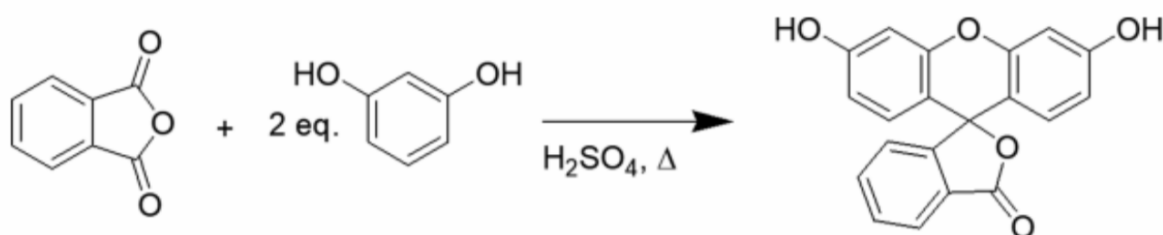


Experiment: Production of a Biphasic Glow Stick through Organic Synthesis, Fluorescence, and Chemiluminescence

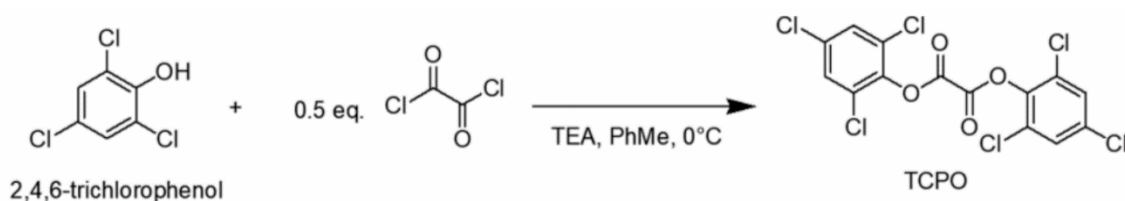
Procedure:

Step 1: Synthesis of fluorescein:



1. Crush 1gm (1 equivalent) of phthalic anhydride and 1.4gm (2 equivalent) of resorcinol using a mortar and pestle. Transfer the resulting powder and a magnetic stirrer bead into a 100 mL beaker.
2. Place the beaker in an oil bath on a magnetic stirrer. Heat the mixture to 150°C - 180°C until it melts.
3. Once melted, carefully add concentrated sulfuric acid to the beaker.
4. Observe the colour change. Once the reaction occurs, remove the beaker from the stirrer and add more concentrated sulfuric acid to dissolve the thick paste.
5. After the paste has melted, add water to the beaker to dilute the mixture.
6. Vacuum filter the solution. The product, fluorescein, will be obtained as a yellowish powder.

Step 2: Synthesis of TCPO:



1. Place a 10 mL round-bottom flask on a stirrer with a magnetic bead inside. Add 500mg (1 equivalent) of 2,4,6-trichlorophenol, cover with a septum, and purge with nitrogen gas.
2. Dissolve the trichlorophenol in dry toluene using a metallic needle. Add 0.35mL (1 equivalent) of triethylamine and stir for 5 minutes under nitrogen conditions.
3. After stirring, add 0.1mL (0.5 equivalent) of oxalyl chloride to the solution while in an ice bath. The solution will solidify.
4. Remove the flask from the stirrer, vacuum filter the solution, and wash the solid with methanol. The product, TCPO, will be a white powder.

Step 3: Glow stick assembly

1. **Fluorescein Solution:** Dissolve a small amount of fluorescein powder in a sodium hydroxide solution to produce a green fluorescent color.
2. **Chemiluminescent Solution:** Prepare a solution of sodium acetate and diphenyl anthracene (1:1) in diethyl phthalate. Add a small amount of TCPO to the chemiluminescent solution. A blue chemiluminescent glow will be observed.
3. **Glow Stick:** Combine the fluorescent and chemiluminescent solutions in a test tube. Observe the glow stick effect in a dark room, where chemiluminescence and fluorescence will be visible.